
Exact Symbolic Marginalization Enables Unseen-Sum Generalization in AddMNIST

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Abstract

Neural-symbolic learning is often motivated by systematic generalization: a model should learn perceptual predicates from data while using symbolic structure to compose them beyond the observed training labels. We test this claim in a controlled ADDMNIST setting where training excludes all pairs in which both digits are high, i.e., both in $\{5, \dots, 9\}$. This restriction removes sums 14–18 from the supervised training labels while preserving the symbolic rule for addition. We compare a direct pair-image CNN against three neural-symbolic models that learn digit predicates and compute pair sums by exact marginalization over the addition table, including two anchor-based variants. With 5,000 weak pair labels, the plain symbolic likelihood model reaches 90.0% IID accuracy and 82.5% OOD high-plus-high accuracy, while the direct CNN reaches 31.8% IID accuracy and 0.3% OOD accuracy. On the truly unseen OOD sums 14–18, the symbolic likelihood model reaches 80.6% accuracy, whereas the direct CNN reaches 0.0%. The proposed anchor-posterior variant improves strongly over the direct baseline but does not beat the simpler symbolic likelihood. These results show that exact symbolic marginalization can extrapolate to labels never observed during training, but only after the neural digit predicates become well grounded.

1 Introduction

Can a model predict labels it never saw? This question is central to neural-symbolic learning. If a system learns visual predicates such as digit identity and then composes them with an exact rule such as addition, it should be able to predict a new sum even when that sum was absent from the supervised labels. A direct neural classifier, by contrast, can learn strong correlations in the observed label space but has no structural reason to assign probability to labels that never appear during training.

Neural-symbolic methods promise this kind of systematic generalization by separating perception from reasoning. Prior systems connect neural predicates with probabilistic logic, differentiable logic, or declarative symbolic programs Manhaeve et al. [2018], van Krieken et al. [2022], Li et al. [2023], Badreddine et al. [2022]. Yet it is still useful to isolate a small, reproducible stress test where the output label gap is explicit. We focus on ADDMNIST because it exposes the exact tension: the input is perceptual, the rule is known, and the training distribution can be edited so that some valid sums never occur as labels.

Our main question is simple. Can a compact neural-symbolic ADDMNIST learner generalize to high-plus-high digit pairs when all such pairs are removed from training? We train on pairs where at least one digit is below 5, and we evaluate out of distribution on pairs where both digits are in $\{5, \dots, 9\}$. This makes sums 14–18 absent from training while keeping them common at test time. Figure 1 shows the comparison: the direct baseline predicts a sum from two images, while the symbolic models predict digit distributions and marginalize over digit pairs that satisfy the addition table.

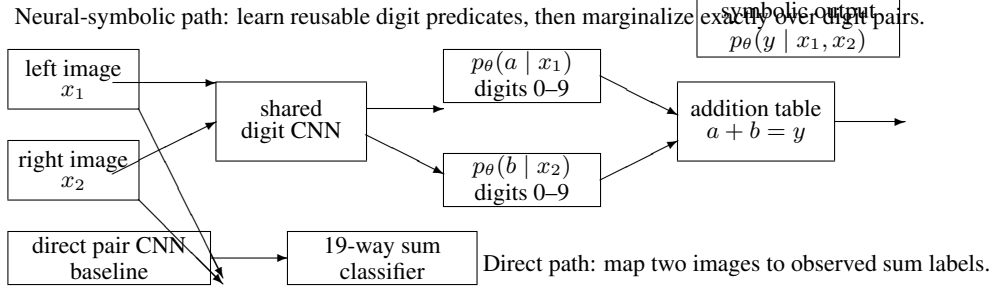


Figure 1: Overview of the comparison. The neural-symbolic models share a digit recognizer across both images and compute the sum distribution by exact marginalization over the fixed addition table. The direct baseline instead trains a 19-class classifier from image pairs to sum labels.

We evaluate four models: a direct pair CNN (DIRECTSUM), a weakly supervised symbolic likelihood model (SYMBOLICSUM), a symbolic model with 10 labeled digit anchors per class (ANCHORSYMBOLIC), and an anchor-posterior variant that adds entropy regularization over the latent digit-pair posterior (ANCHORPOSTERIOR). The result is clear at the larger pair-label budget. With 5,000 pair labels, SYMBOLICSUM reaches 82.5% OOD high-plus-high accuracy, compared with 0.3% for DIRECTSUM. On sums 14–18, which never appear as supervised labels, SYMBOLICSUM reaches 80.6%, while DIRECTSUM reaches 0.0%.

Our contributions are:

- We define a controlled ADDMNIST extrapolation protocol that removes all high-plus-high training pairs and therefore removes sums 14–18 from the supervised labels.
- We implement exact symbolic marginalization over learned digit predicates and compare it against a direct neural sum classifier under matched budgets and seeds.
- We evaluate anchor and posterior-entropy ablations, showing that these additions improve over the direct baseline but do not beat the plain symbolic likelihood in this run.
- We connect OOD accuracy to digit grounding, paired seed-level tests, and bootstrap confidence intervals over 192,000 per-example predictions.

Paper organization. Section 2 positions the experiment relative to prior neural-symbolic systems. Section 3 defines the task, models, and evaluation protocol. Section 4 reports the main results, unseen-label analysis, grounding diagnostics, and statistical tests. Section 5 discusses interpretation, limitations, and next steps.

2 Related Work

Probabilistic neural-symbolic inference. DeepProbLog integrates neural predicates into ProbLog so neural outputs can serve as probabilistic facts inside symbolic programs Manhaeve et al. [2018]. It helped establish MNIST arithmetic as a benchmark for weakly supervised perception plus logic. A-NeSI keeps discrete symbolic semantics while using approximate inference networks to scale beyond the small cases where exact probabilistic inference is practical van Krieken et al. [2022]. Our experiment follows this line in spirit, but it uses a deliberately small exact marginalization layer rather than a full probabilistic programming stack. This choice lets us isolate the core question of unseen output-label extrapolation.

Declarative and differentiable logic. Scallop provides a Datalog-like language for neurosymbolic programs with provenance semantics and differentiable reasoning Li et al. [2023]. Logic Tensor Networks ground first-order logic into differentiable objectives, and LTNTorch makes that approach available in PyTorch Badreddine et al. [2022], Carraro et al. [2024]. These systems are broader than our implementation: they support many rules, relations, and reasoning patterns. We instead use one exact rule, addition, and test whether the symbolic computation alone changes behavior when the label support shifts.

Relational and concept-level generalization. Neural Logic Machines learn tensorized logical operations over objects and relations, targeting length and size generalization in relational domains Dong et al. [2019]. The Neuro-Symbolic Concept Learner connects visual perception, semantic parsing, and symbolic program execution for visual question answering Mao et al. [2019]; later concept-intervention work studies whether symbolic concepts can make corrections more meaningful Stammer et al. [2021]. These papers address richer reasoning settings. Our study is narrower, but the narrowness is useful: digit grounding can be measured directly, and every OOD sum has an exact symbolic explanation.

AddMNIST as a compositional probe. MNIST arithmetic remains useful because it has a clean latent structure: the image labels are not the target labels, but the target follows from a known rule LeCun et al. [1998], Manhaeve et al. [2018]. Many ADDMNIST evaluations report in-distribution accuracy or use larger arithmetic variants. We focus on an explicit support gap: high-plus-high pairs are removed from training, so the learner never receives supervised examples for sums 14–18. This makes the evaluation less about memorizing arithmetic labels and more about composing learned digit predicates.

3 Methodology

3.1 Task Formulation

Each example contains two MNIST images (x_1, x_2) with latent digits $(d_1, d_2) \in \{0, \dots, 9\}^2$ and target sum $y = d_1 + d_2 \in \{0, \dots, 18\}$. The training set contains weak pair labels y but, except for the anchor variants, does not reveal d_1 or d_2 . The central design choice is the train/test split. Training pairs exclude the high-plus-high region,

$$\{(d_1, d_2) : d_1 \geq 5 \text{ and } d_2 \geq 5\}, \quad (1)$$

while the OOD split contains only that region. As a consequence, sums 14–18 never appear as supervised training labels, although they are valid outputs of the addition rule.

3.2 Data and Splits

We use the local MNIST dataset, with 60,000 training images and 10,000 test images LeCun et al. [1998]. Images are normalized with the standard MNIST mean and standard deviation. For each seed, we sample restricted training pairs for budgets of 1,000 and 5,000 pair labels. We evaluate on 4,000 IID restricted pairs and 4,000 OOD high-plus-high pairs per run. Across three seeds and four methods, the saved predictions contain 192,000 per-example pair predictions.

3.3 Models

DIRECTSUM. The direct baseline is a CNN that concatenates the two digit images as two input channels and predicts one of 19 sum labels with cross-entropy. The classifier has output logits for every sum, including 14–18, but labels 14–18 are absent from the restricted training set.

SYMBOLICSUM. The symbolic likelihood model uses a shared digit CNN f_θ to produce digit distributions for each image. It then computes the probability of a sum by exact marginalization:

$$p_\theta(y \mid x_1, x_2) = \sum_{a=0}^9 \sum_{b=0}^9 \mathbf{1}[a + b = y] p_\theta(a \mid x_1) p_\theta(b \mid x_2). \quad (2)$$

The model is trained with negative log likelihood on the observed pair sum. This objective has no learned rule parameters; the only learned part is digit perception.

ANCHORSYMBOLIC. The anchor variant adds a small balanced digit-supervision set with 10 labeled examples per digit class. The loss combines the pair-sum negative log likelihood with an anchor cross-entropy term weighted by $\lambda = 0.5$.

ANCHORPOSTERIOR. The posterior variant adds entropy regularization over the latent digit-pair posterior induced by the observed sum. For a pair and sum label y , the posterior is

$$q_\theta(a, b \mid x_1, x_2, y) = \frac{\mathbf{1}[a + b = y] p_\theta(a \mid x_1) p_\theta(b \mid x_2)}{p_\theta(y \mid x_1, x_2)}. \quad (3)$$

Pair labels	Method	IID acc. (%)	OOD acc. (%)	OOD 95% CI (%)	Digit acc. (%)
1,000	DIRECTSUM	14.3	1.4	[1.2, 1.6]	–
1,000	SYMBOLICSUM	17.3	0.0	[0.0, 0.0]	29.4
1,000	ANCHORSYMBOLIC	19.9	0.0	[0.0, 0.0]	34.1
1,000	ANCHORPOSTERIOR	18.9	0.0	[0.0, 0.0]	33.4
5,000	DIRECTSUM	31.8	0.3	[0.2, 0.4]	–
5,000	SYMBOLICSUM	90.0	82.5	[81.8, 83.2]	93.8
5,000	ANCHORSYMBOLIC	87.7	73.2	[72.4, 74.0]	91.8
5,000	ANCHORPOSTERIOR	88.0	74.1	[73.3, 74.8]	92.0

Table 1: Main AddMNIST results across budgets. Higher is better for all accuracy columns. Best result within each pair-label budget and metric is in **bold**. OOD confidence intervals are bootstrap 95% intervals over 12,000 predictions per method-budget condition.

Method	Seen OOD sums 10–13 (%)	Examples	Unseen sums 14–18 (%)	Examples
DIRECTSUM	0.9	4,432	0.0	7,568
SYMBOLICSUM	85.7	4,432	80.6	7,568
ANCHORSYMBOLIC	79.2	4,432	69.7	7,568
ANCHORPOSTERIOR	80.6	4,432	70.3	7,568

Table 2: OOD high-plus-high accuracy at the 5,000-label budget, split by whether the target sum label appeared in training. Sums 14–18 never appear as supervised training labels because training excludes high-plus-high pairs.

The training loss adds $\beta H(q_\theta)$ with $\beta = 0.03$, so minimizing the loss encourages sharper latent assignments among digit pairs consistent with the observed sum. This model also uses the anchor loss.

3.4 Training and Evaluation

All models are trained for 15 epochs with batch size 512, AdamW, learning rate 10^{-3} , and weight decay 10^{-4} . We run seeds 11, 23, and 37. Experiments use PyTorch 2.12.1 on an NVIDIA RTX A6000 with mixed precision enabled.

We report sum accuracy on the IID restricted split and on the OOD high-plus-high split. For symbolic models, we also report digit accuracy on the MNIST test split to check whether the learned predicates align with true digits. We compute macro-F1, negative log likelihood, mean confidence, and training time in the result files, and we emphasize accuracy because it directly measures the extrapolation claim. For the primary OOD metric, we use paired seed-level t -tests and Cohen’s d_z effect sizes, plus bootstrap 95% confidence intervals over per-example predictions.

4 Results

4.1 Main Accuracy Results

Figure 2 and table 1 show the main result. At 1,000 pair labels, none of the symbolic models learns a reliable enough digit grounding to solve the high-plus-high split. DIRECTSUM obtains 1.4% OOD accuracy, while all symbolic variants obtain 0.0%. This low-budget result is important: symbolic structure does not remove the need to learn usable predicates.

At 5,000 pair labels, the behavior changes sharply. SYMBOLICSUM reaches 90.0% IID accuracy and 82.5% OOD high-plus-high accuracy. DIRECTSUM improves to 31.8% IID accuracy but remains at 0.3% OOD accuracy. The absolute OOD gap between SYMBOLICSUM and DIRECTSUM is 82.2 percentage points. The anchor variants also strongly beat the direct baseline, reaching 73.2% and 74.1% OOD accuracy, but they do not exceed the plain symbolic likelihood.

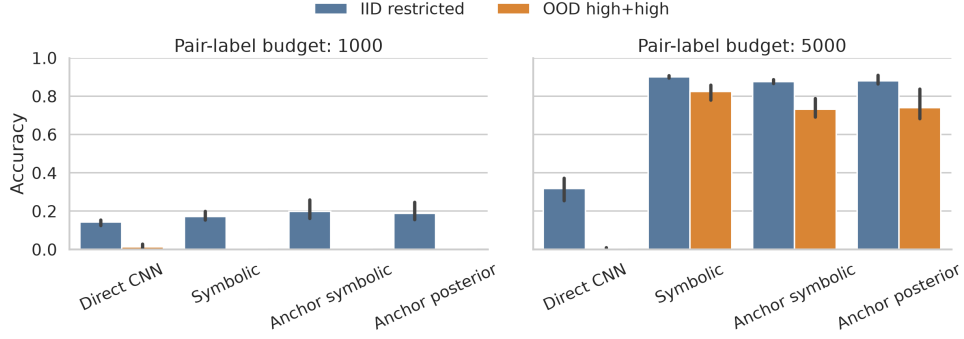


Figure 2: Accuracy by method and pair-label budget. The 5,000-label setting separates the direct classifier from the neural-symbolic models: exact symbolic marginalization enables high OOD accuracy once digit predicates are grounded.

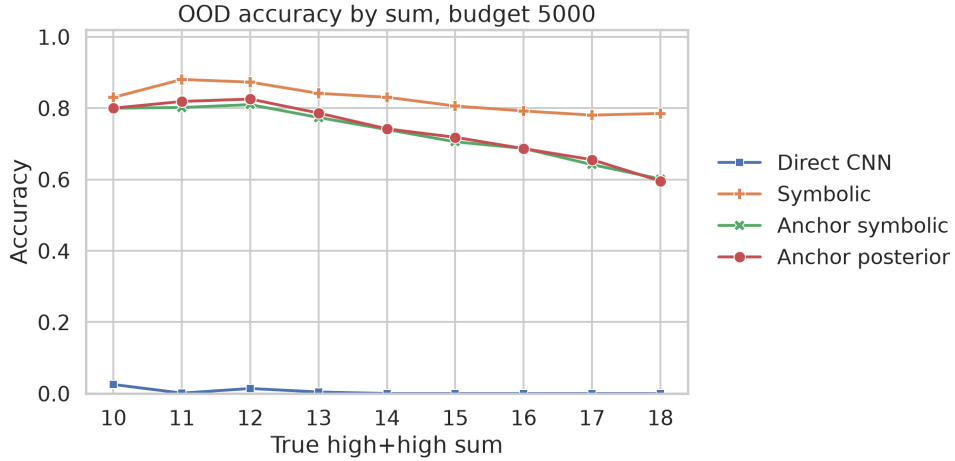


Figure 3: OOD accuracy by true sum. The symbolic models recover high-plus-high sums across the OOD range at the 5,000-label budget, including sums 14–18 that never appear as training labels.

4.2 Extrapolation to Unseen Sum Labels

The main OOD split contains both seen and unseen sum labels. Sums 10–13 can occur during restricted training through low-plus-high pairs, while sums 14–18 cannot occur because both digits must be high. Table 2 separates these regions for the 5,000-label budget. SYMBOLICSUM reaches 85.7% accuracy on seen OOD sums and 80.6% accuracy on unseen sums 14–18. DIRECTSUM reaches 0.9% on seen OOD sums and 0.0% on unseen sums. The direct model has output logits for the unseen labels, but the training objective never rewards those labels, so it has no effective route to the held-out sums.

4.3 Digit Grounding

The OOD gains coincide with strong digit grounding. At 1,000 pair labels, symbolic models reach only 29.4–34.1% digit accuracy, and OOD accuracy remains at zero. At 5,000 pair labels, digit accuracy rises above 91% for all symbolic variants. The strongest OOD model, SYMBOLICSUM, also has the strongest digit grounding at 93.8% mean digit accuracy. This supports the intended interpretation: the model is not merely fitting aggregate sum labels; it learns reusable digit predicates that the addition table can compose.

Method	1,000 labels digit acc. (%)	5,000 labels digit acc. (%)
SYMBOLICSUM	29.4 $\pm$ 4.8	93.8 $\pm$ 0.8
ANCHORSYMBOLIC	34.1 $\pm$ 6.1	91.8 $\pm$ 1.1
ANCHORPOSTERIOR	33.4 $\pm$ 5.9	92.0 $\pm$ 2.2

Table 3: Digit grounding accuracy for symbolic models on the MNIST test split. Values report mean and standard deviation over three seeds.

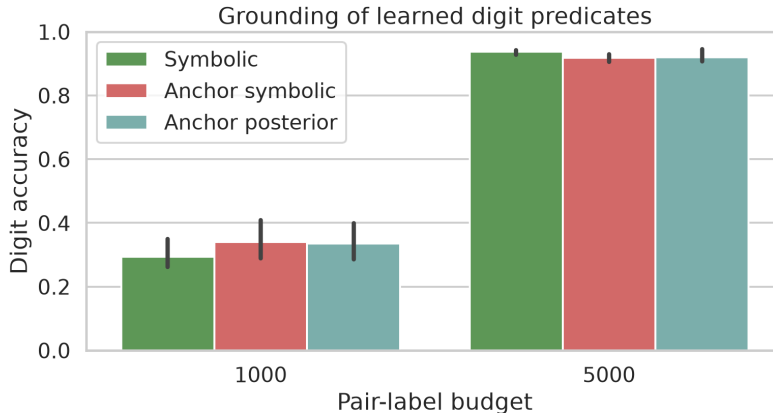


Figure 4: Digit grounding accuracy for symbolic models. OOD sum generalization appears only when the learned digit predicates become accurate enough to support symbolic composition.

4.4 Statistical Comparisons

Table 4 reports paired seed-level tests for the 5,000-label OOD setting. The symbolic likelihood model improves over DIRECTSUM by 82.2 percentage points, with Cohen’s $d_z = 20.75$ and $p = 0.0008$ across three matched seeds. The anchor variants also show large gains over DIRECTSUM. Because there are only three seeds, we treat the p -values as descriptive rather than definitive. The bootstrap intervals over per-example predictions tell the same story: SYMBOLICSUM has an OOD interval of [81.8, 83.2]%, while DIRECTSUM has [0.2, 0.4]%.

4.5 Ablation Summary

The planned anchor-posterior modification is useful as an ablation but not as the best-performing method. At 5,000 labels, ANCHORPOSTERIOR reaches 74.1% OOD accuracy, slightly above ANCHORSYMBOLIC at 73.2%, but both trail SYMBOLICSUM at 82.5%. The anchor and entropy terms may over-weight early latent assignments or interfere with the weak pair-label signal. The result is a negative finding for the proposed modification and a positive finding for the simpler symbolic likelihood.

5 Discussion

Symbolic composition works after perception is grounded. The experiment supports the central neural-symbolic hypothesis in this controlled setting. When the digit CNN learns accurate predicates, exact marginalization over the addition table produces valid sums outside the supervised label support. This is why SYMBOLICSUM can reach 80.6% accuracy on sums 14–18 even though those labels never appear during training. The low-budget condition also clarifies the boundary of the claim: symbolic structure cannot compensate for poorly grounded predicates.

The simplest symbolic likelihood is the strongest model. We expected the anchors and posterior entropy to stabilize latent digit assignments. They do help relative to the direct baseline at 5,000 labels, but they do not improve over SYMBOLICSUM. One explanation is that the sum likelihood

Method A	Method B	Mean A (%)	Mean B (%)	Diff. (%)	Cohen’s d_z	p -value
ANCHORPOSTERIOR	DIRECTSUM	74.1	0.3	73.8	9.33	0.0038
ANCHORSYMBOLIC	DIRECTSUM	73.2	0.3	72.9	16.67	0.0012
SYMBOLICSUM	DIRECTSUM	82.5	0.3	82.2	20.75	0.0008

Table 4: Paired seed-level tests for OOD high-plus-high accuracy at the 5,000-label budget. Tests use three matched seeds and should be read as descriptive because the seed count is small.

already provides enough signal once 5,000 pair labels are available, while the anchor and entropy terms add constraints that are too strong early in training. A better variant may need a schedule that delays posterior sharpening or reduces anchor weight after the digit classes become identifiable.

The direct baseline fails for the intended reason. DIRECTSUM is not a universal neural baseline; it is a direct classifier under a deliberately harsh label-support shift. It can improve on IID restricted pairs, but the missing labels 14–18 receive no positive supervised examples. The point of the comparison is therefore not that direct CNNs cannot learn MNIST arithmetic in general. The point is that a direct label map and a symbolic composition model make different extrapolation commitments when the label support changes.

Limitations. ADDMNIST is synthetic and much simpler than language reasoning, visual question answering, or knowledge-graph reasoning. We use an exact known rule, so the experiment does not test rule induction. The OOD split changes digit-pair composition but still uses the same MNIST image distribution, so it is not a visual domain-shift benchmark. We also run only three seeds. The OOD effects at 5,000 labels are large and consistent, but more seeds and broader hyperparameter sweeps would give a stronger statistical picture.

Broader implications. The results argue for treating exact symbolic marginalization as a strong baseline whenever the rule is known and the latent predicate space is small. It is easy to overlook this baseline in favor of heavier neural-symbolic frameworks, but this experiment shows that the simple version can already deliver the desired label extrapolation. The next challenge is to preserve this advantage as the predicate space grows, the rule becomes approximate, or perception shifts away from the training domain.

6 Conclusion

We studied whether neural-symbolic learning can generalize to ADDMNIST sums that never appear as supervised training labels. The answer is yes in the 5,000-label setting, but only after the digit predicates are well grounded. The plain symbolic likelihood model reaches 82.5% OOD high-plus-high accuracy and 80.6% accuracy on unseen sums 14–18, while the direct pair CNN remains near zero OOD accuracy.

The main practical lesson is that exact symbolic marginalization is a powerful and simple mechanism for compositional label extrapolation. The anchor-posterior variant is a useful negative result: additional grounding and sharpening terms did not improve over the weak symbolic likelihood in this run. Future work should tune anchor and posterior schedules, extend the protocol to three-digit arithmetic where exact marginalization becomes more expensive, compare against Scallop, Deep-ProbLog, or A-NeSI implementations, and add noisy or shifted MNIST variants to test whether the compositional gain survives perceptual domain shift.

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